

DIGITAL MONEY For Digital India



Manu Prasad is M.Tech in VLSI and embedded systems, and is currently working as assistant professor at AWH Engineering College, Kerala. His interests include VLSI, EDA tools, MATLAB and LaTeX

On November 8, 2016, at 20:15, Prime Minister of India Narendra Modi addressed the nation through an unscheduled television speech. He declared that from November 9, 2016, the government of India would cease the usage of all 500- and 1000-rupee notes as legal tender, and instead new 500- and 2000-rupee notes would be available for exchange. The government claimed that this was being done to stop terrorism funding, crack down on black money, and reduce corruption and smuggling in India. However, after the announcement, banks and ATMs in the whole country faced severe shortages of currency.

The government of India is now trying to implement a cashless society. After the demonetisation move, several initiatives were taken to further encourage cashless transactions. Almost 300 per cent increase in digital payment activities has been observed since early November 2016.

In this article we present some of the many digital alternatives for making payments.

Debit and credit cards

A debit card is used to withdraw money from ATMs and to make purchases online and offline. The money is debited directly from your bank account. The card has a black magnetic strip on the back or a chip on one side, in which your account information and other details are embedded.

For making payments using a debit card, the merchant uses a point of sale (POS) machine to swipe or insert the card. Steps to be followed in this method are as follows:

1. While accepting a pay-

ment, the merchant submits the details after swiping or inserting the debit card on the POS machine.

2. You provide your personal identification number (PIN) through the machine.

3. After submission, the device communicates with your bank and deducts the amount from your account.

4. Two receipts are printed, one for you and the other for the merchant on the POS machine after successful payment.

5. An SMS from the bank confirming the transaction is sent to your registered mobile number.

In case of a credit card, the payment amount is not deducted while using the card; the payment is made later by the user.

Most online shopping websites support the use of debit and credit cards. Here, shopping vendors direct you to the selected bank payment portals, where you enter your debit or credit card details like card number (16-digit number provided on the card), validity date and CVV number (last three digits provided on the back of the card next to your signature). Then, you have to provide a one-time password sent by your bank to your phone or your Internet password for netbanking. A successful transaction message page is displayed after the payment is done.

RuPay is an Indian domestic card launched by National Payment Corp. of India. It facilitates electronic payments from all Indian banks. The procedure for using RuPay is the same as that of other debit cards.

Mobile/digital wallets

A mobile wallet is a virtual wallet where you can preload a certain amount of money from your bank account, and use this amount as cash. For example, if you go to a coffee shop and the coffee shop supports, say, PayTM wallet, you can pay your bill using the same.

There are four types of mobile wallets in India, namely, open, semi-open, closed and semi-closed. Open wallets allow you to buy goods and services, withdraw cash at

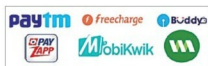


Fig. 1: Some popular mobile wallets in India



Fig. 2: Adding a beneficiary account